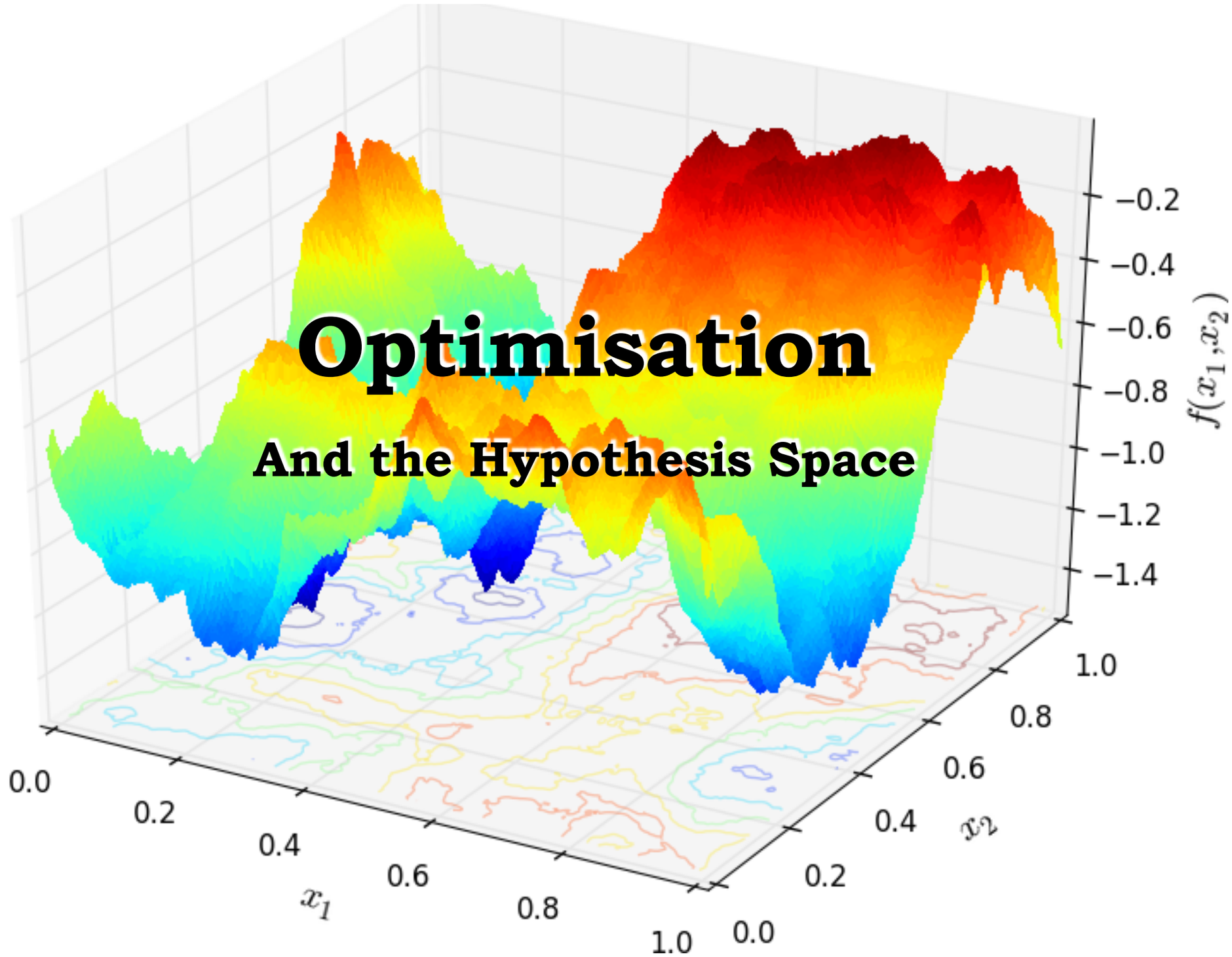
Optimisation

And the Hypothesis Space



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Optimisation - The Heart of AI

Perhaps a more functional metaphor is that optimisation is the ***engine*** - "what makes it tick".

We all have a general understanding of what we mean by optimisation

optimization noun

Save Word

op·ti·mi·za·tion | \ ˌäp-tə-mə-ˈzā-shən \

Definition of *optimization*

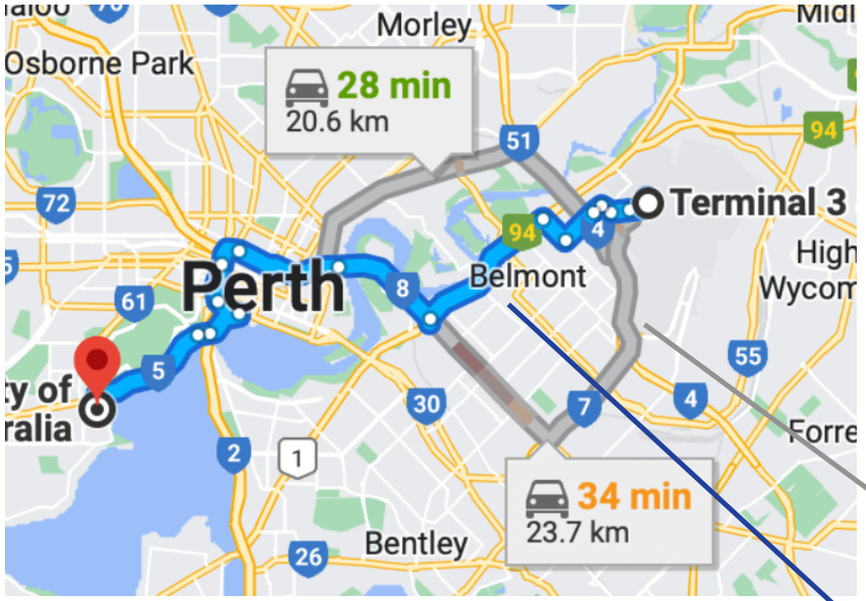
: an act, process, or methodology of making something (such as a design, system, or decision) as fully perfect, functional, or effective as possible

specifically **:** the mathematical procedures (such as finding the maximum of a function) involved in this

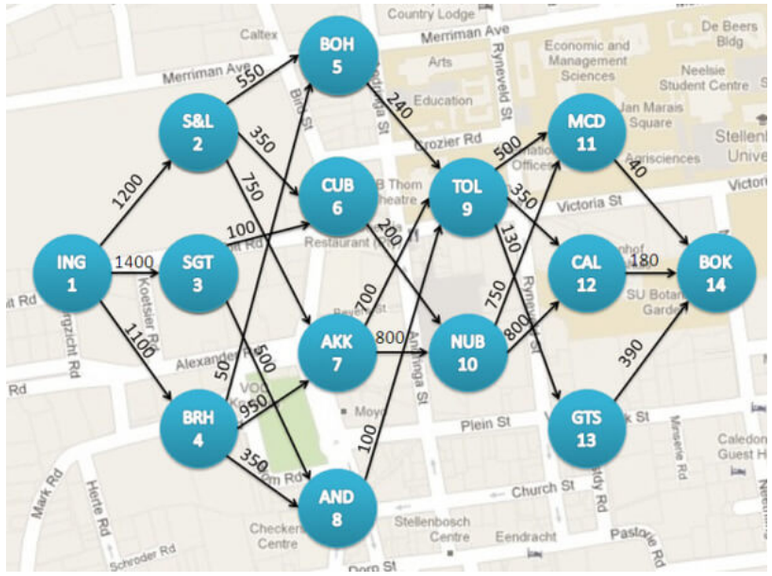
In this section we are going to be more specific

- dissect the key components that underpin the many faces of optimisation
- introduce the language/terms we will use in this unit

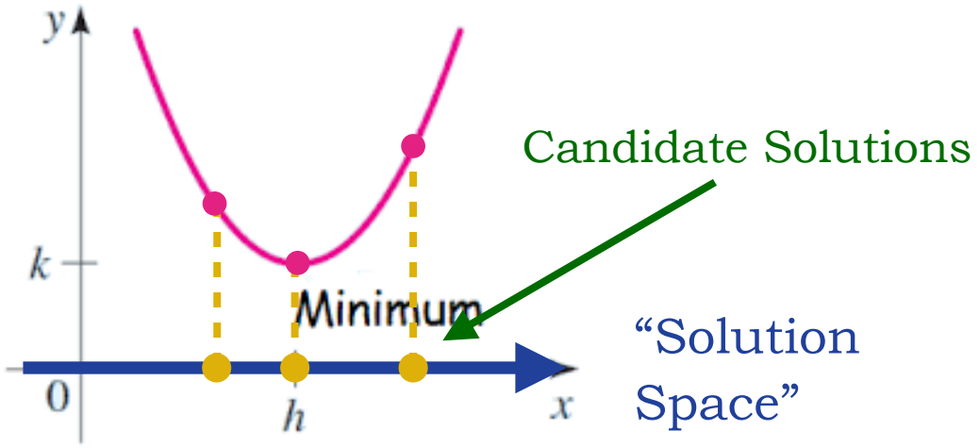
Thinking about Optimisation (Taster)



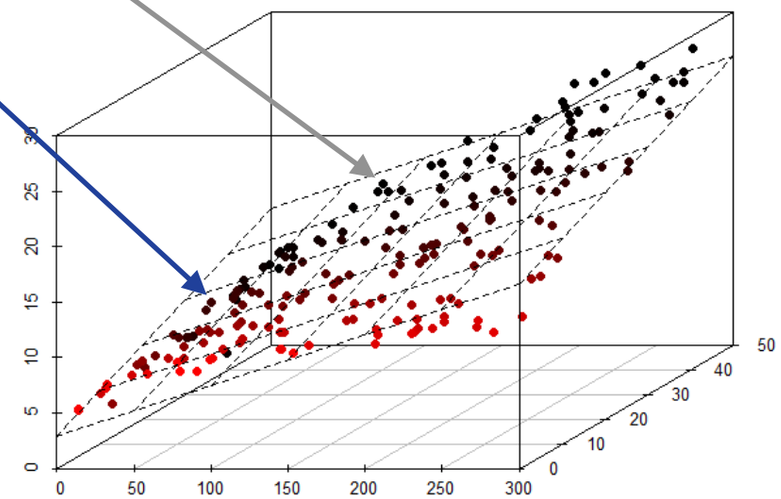
Problem



Abstract Representation

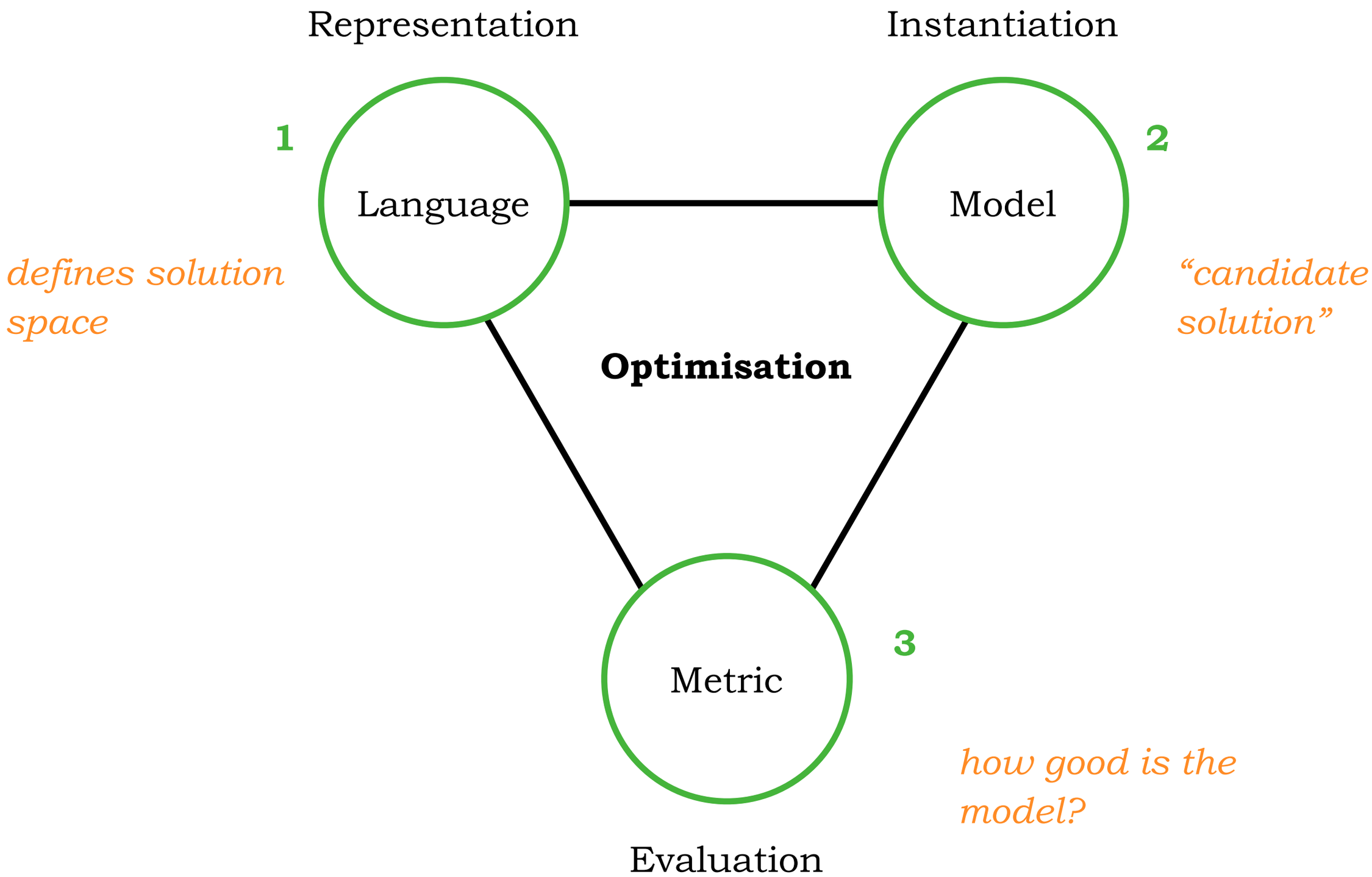


Solution (v. simple problem)



(Candidate) Solutions

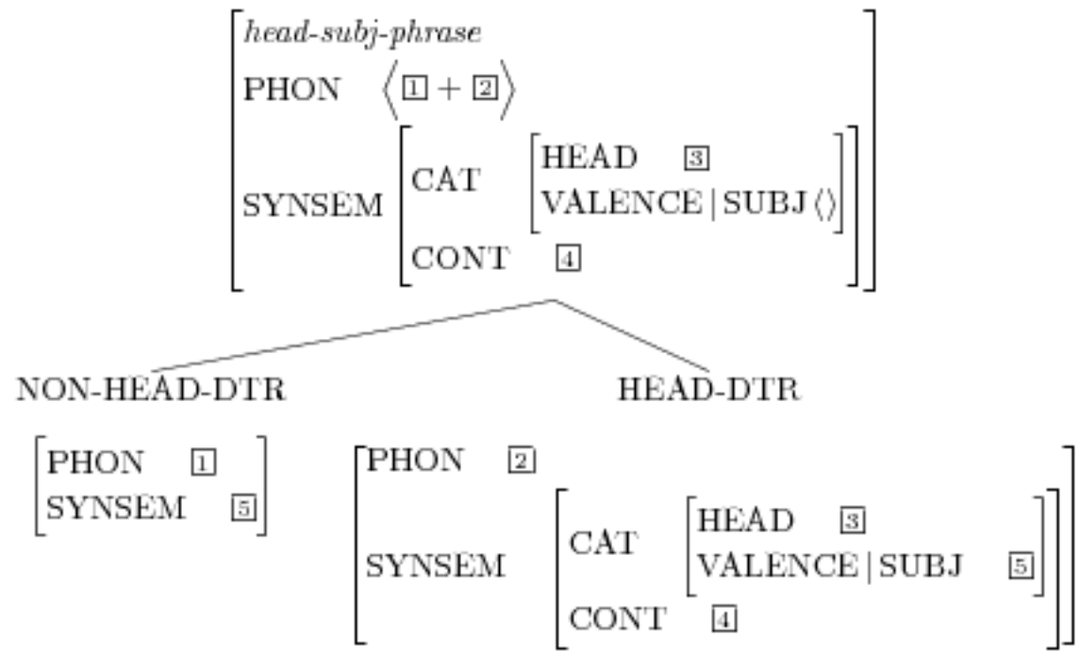
Three Key Ingredients



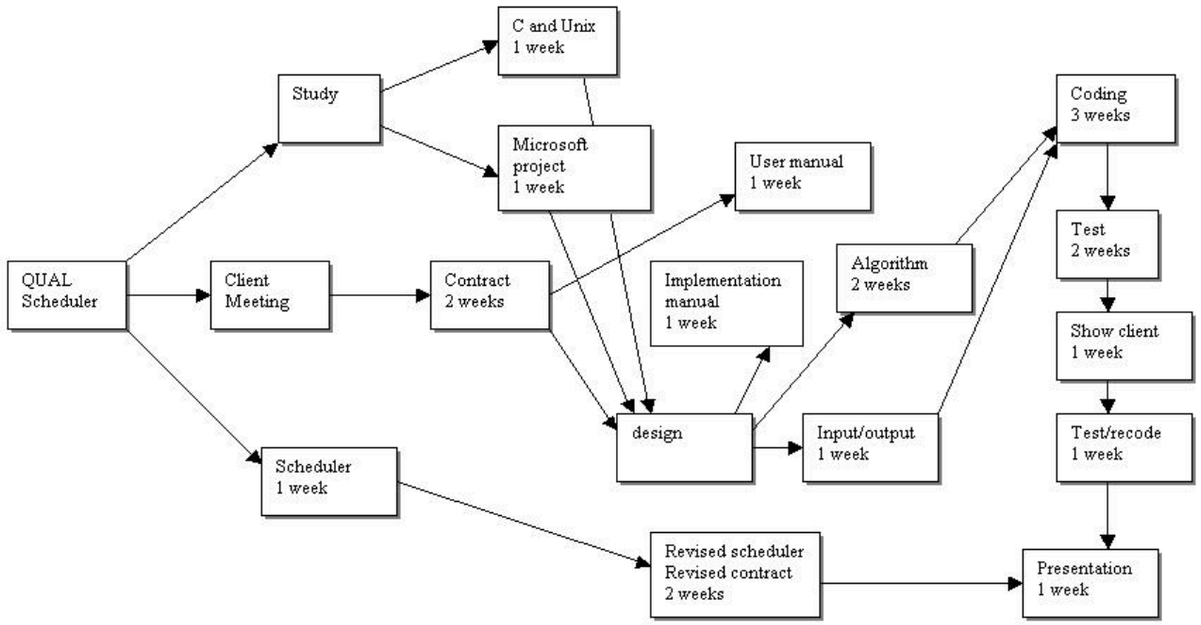
1. Language (Representation)

- mathematical equations
- natural languages
- grammars
- logics
- finite automata/finite-state machines
- computer programs
- logic programs
- Gantt charts
- PERT charts
- simulation languages
- popsticks and glue
- …

HPSG



PERT CHART



Language

if you can't describe it - you can't model it!

Example

- Enormous amount of work on grammars to *describe (explain/ parse/ generate/ model)* natural languages
  - CFGs, LFGs, PSGs, GPSGs, HPSGs, TGGs,...

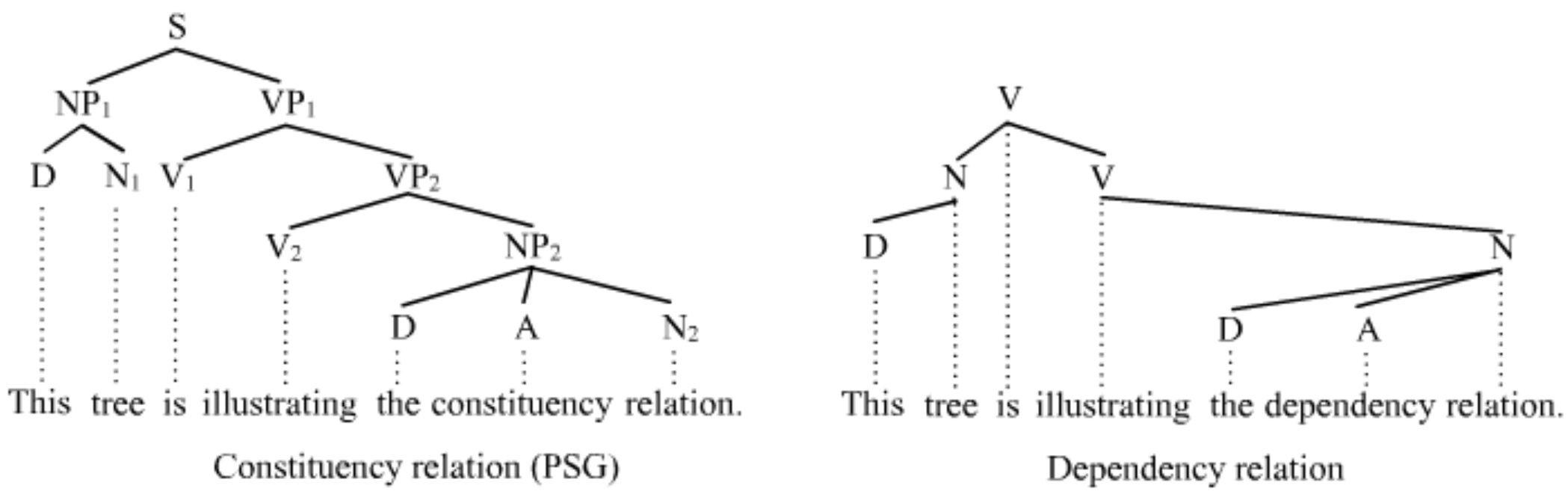


Image: Tjo3ya, Wikipedia

Generation vs Parsing

Example: *Phrase Structure Grammar* (PSG)

A Simple Set of Phrase Structure Rules

S ⟶ NP +VP

NP ⟶ art + (adj*) + N

VP ⟶ V + NP + (PP*)

PP ⟶ Prep + NP

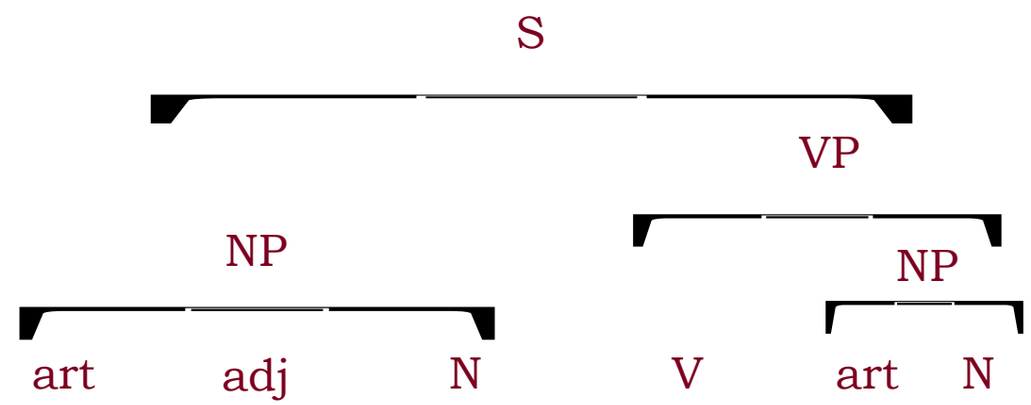
N ⟶ sailor, cat, horse, bridge,

V ⟶ saluted, kissed, fried

adj ⟶ drunken, puzzled, gregarious

art ⟶ a, the

prep ⟶ on, under



the drunken sailor saluted the cat

Parsing

Which of the following sentences are in the space (set) of sentences generated by our grammar? (ie. Are valid/accepted sentences)

1. the drunken gregarious sailor kissed the puzzled cat
2. the bridge saluted the horse under the cat
3. the bridge saluted the bridge under the drunken bridge
4. a sailor fried a cat under a bridge under a bridge
5. fried horse on the bridge
6. the fried horse on the bridge

Test vs Generate

Note that we don't have to *generate* the whole space to answer these questions!

- **Test**
  - we have a test (in this case called parsing) for whether a construct is valid (in the set of sentences)
  - finite time
- **Generate**
  - we can also generate members of the set, but…
  - infinite time

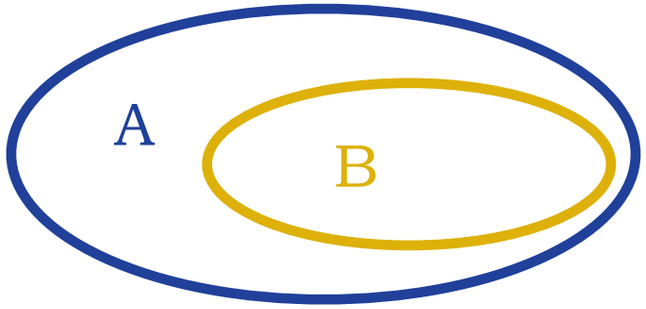
Usually *testing is much cheaper* (eg "polynomial time") *than generating* (we'll return to this)!

Expressiveness of a Language

if you can't describe it - you can't model it!

We will say that language A is ***more expressive*** (or "*richer*") than language B if

- everything that can be described in language B can also be described in language A (i.e. A *subsumes* B)
- something can be described in language A that cannot be described in language B
  (B *does not subsume* A)



- Examples
  - the class of polynomial functions is more expressive than the class of linear functions

$$y = c_n x^n + c_{n-1} x^{n-1} + \cdots + \boxed{c_1 x + c_0} \qquad \text{(polynomial)}$$

$$y = \boxed{ax + b} \qquad \text{(linear)}$$

Language

if you can't describe it - you can't model it!

Example (cf. CITS2211 Discrete Structures)

- The **Chomsky Hierarchy**

Grammar	Language	Automaton	Rules
Type-0	Recursively enumerable	Turing Machine	$\alpha \longrightarrow \beta$
Type-1	Context sensitive	Linear bounded Non-deterministic Turing Machine	$\alpha A \beta \longrightarrow \alpha \gamma \beta$
Type-2	Context Free	Non-deterministic Pushdown Automaton	$A \longrightarrow \gamma$
Type-3	Regular	Finite State Automaton	$A \longrightarrow a$ and $A \longrightarrow aB$

Complexity

Type 3 $\subset$ Type 2 $\subset$ Type 1 $\subset$ Type 0

Expressiveness of a Language

- Examples
  - the class of context-free languages is more expressive than the class of regular languages

$$A \rightarrow \gamma \qquad \text{(context-free)}$$

eg. Contains language $a^n b^n$

$$A \rightarrow a,\ A \rightarrow aB \qquad \text{(regular)}$$

Doesn't contain $a^n b^n$

    - a, terminal
    - A, B, non-terminals
    - γ, string of terminals and/or non-terminals
  - first-order logic (predicate calculus) is richer than propositional logic

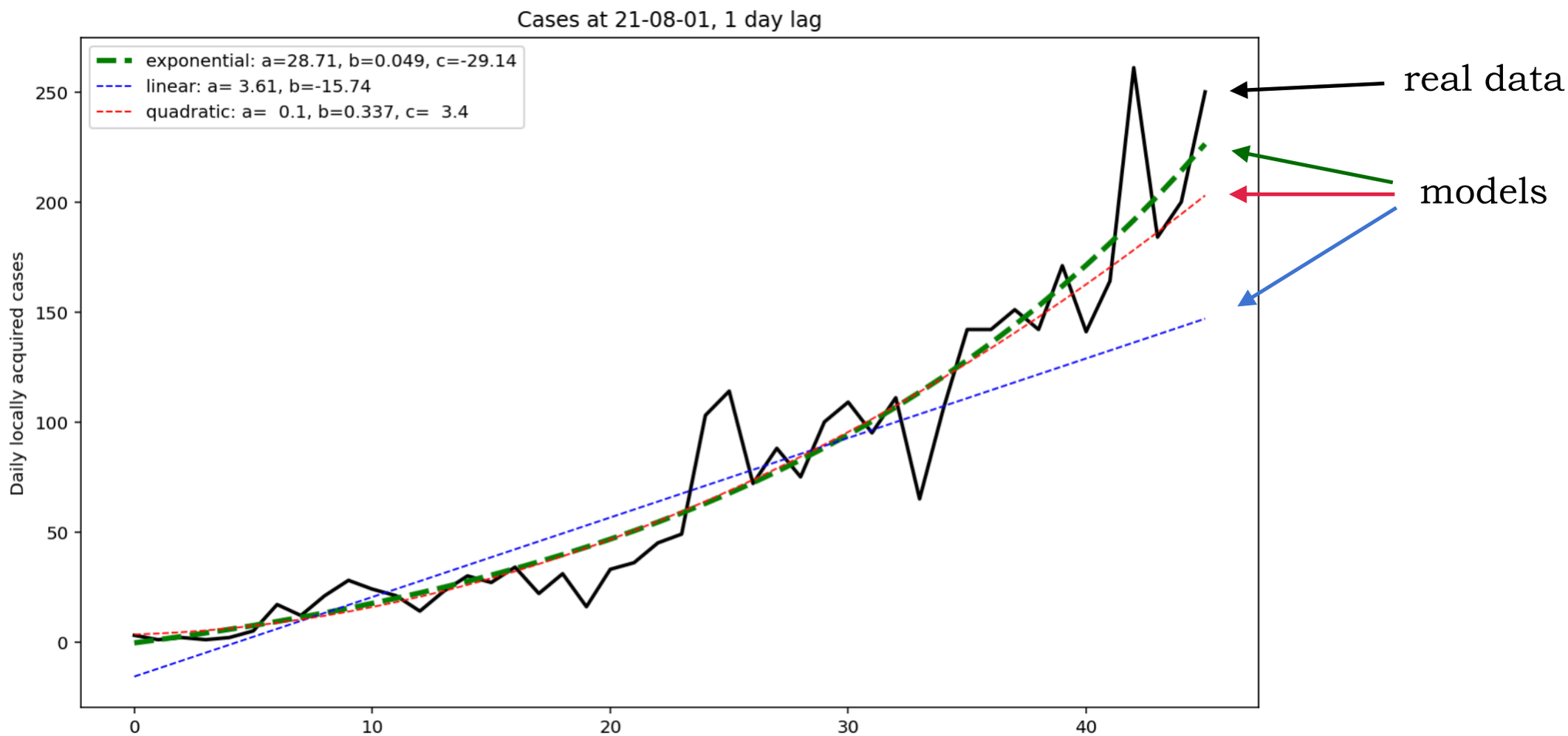
John_Wayne_is_mortal (propositional)

$$\forall x\ \text{man}(x) \implies \text{mortal}(x) \qquad \text{(first-order)}$$

2. Model (Instantiation)

What is it?

- something that can be described in the selected language
- an ***instance*** of all the possible things that can be described in the language
- may be *executable, parameterised, generative, or canonical form*



Model = Hypothesis

Relationship to "real" world

- an *approximation* to the real world (process, function, structure,…)
- an *abstraction* of the real world
- an ***hypothesis*** *that attempts to describe how the real world works* (or could work)
- a ***candidate solution*** (in the context of optimisation)

Everything in the real world is a function!?

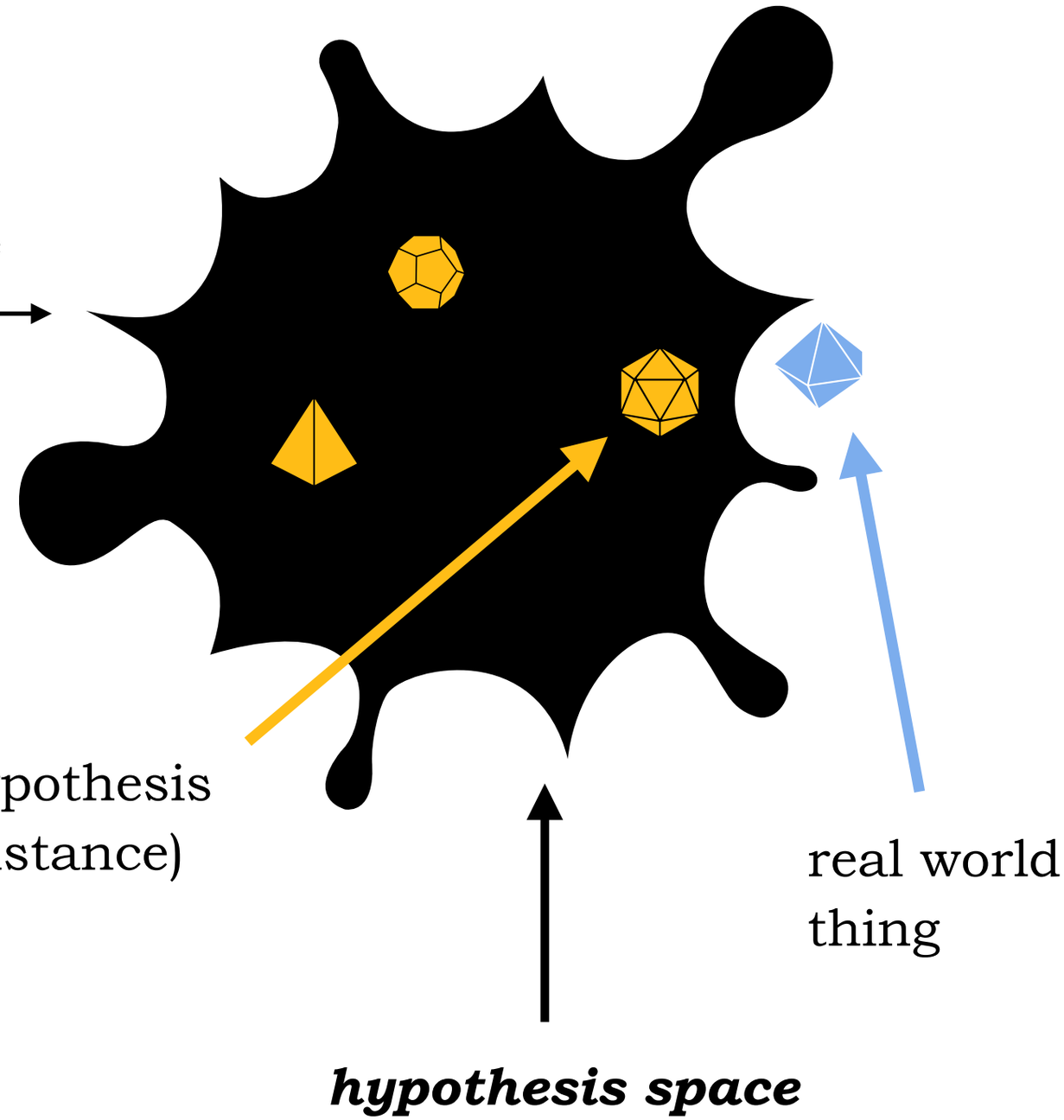
➡ So, *a model/hypothesis* is *an attempt to describe/explain* some *function*, within the *chosen language*

(Note: By "real" we don't necessarily mean "physical".)

Hypothesis Space (Solution Space)

Language
(generative)

all the things you can describe

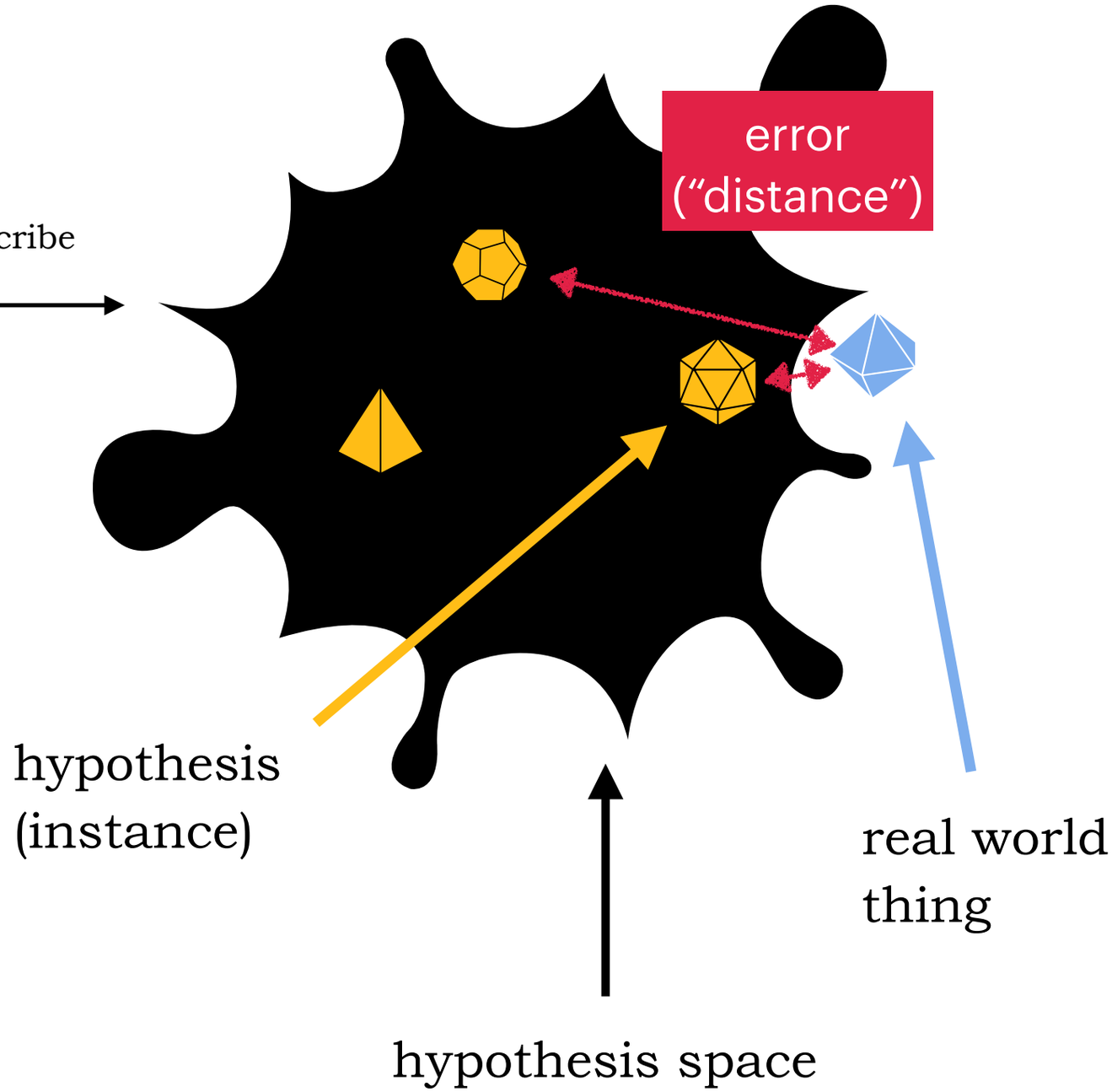


3. Metric (Evaluation)

How "***good***" is your hypothesis?

Language
(generative)

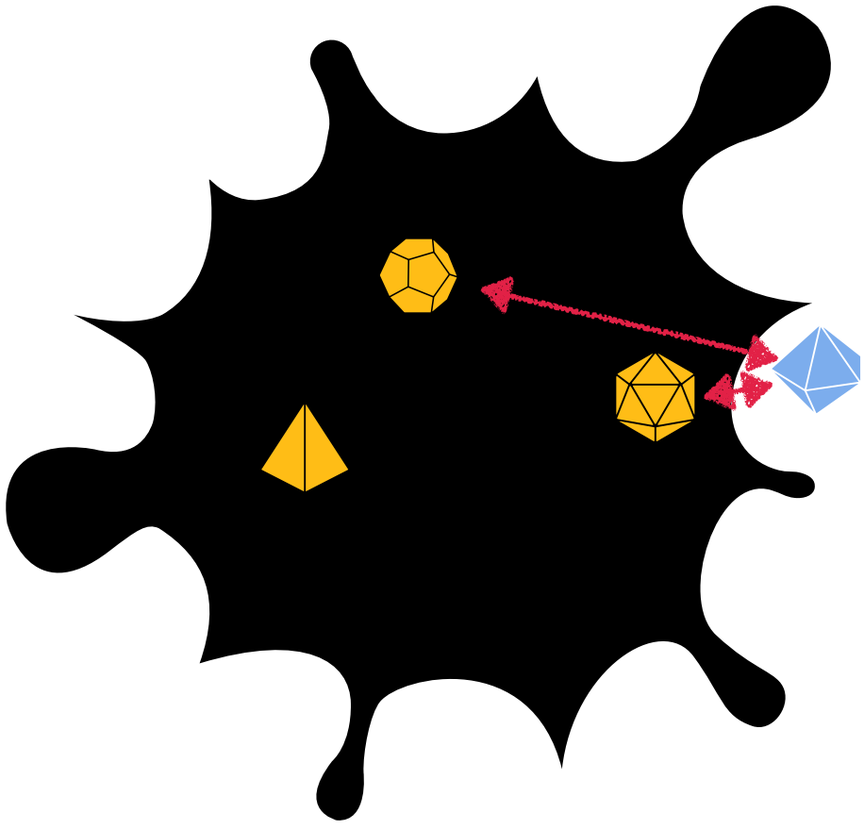
all the things you can describe



- How much is it "like" the real thing?
- How "close" is it to the target?

Metric

- Need some way to ***evaluate*** how "good" an hypothesis is
- There are many ways to describe/ measure "closeness"
  - Usually we are happy if we can determine ***relative*** *closeness*
    - doesn't need to be meaningful in an absolute sense, only relative to another
    - which of these hypotheses is *better*?
- Sometimes we don't know so much about the "real" thing
  - can just assume its infinitely 'good'
  - seek the ***best*** hypothesis



Metrics & Evaluation

- Come under many names…
  - *error function/ metric*
  - *cost function*
  - *fitness function*
  - *objective function*
  - *penalty function*
  - *utility function*
- (Usually) function from hypothesis space onto $\mathbb{R}$

  $f : H \to \mathbb{R}$

  - (sometimes we have multiple objectives)

Features

- Just as the choice of language determines what hypotheses are available to choose from, it also impacts what ***features*** are available to compare
- Example
  - In 2016 the *Australian Bureau of Statistics* (ABS) declared the "Typical Australian" (let's call her the model of an Australian?) was Claire

  - Claire, we were told, has the following features…

Features

The 'Typical' Australian	
Median Age	38
Sex (Mode)	Female
Country of Birth of Person (Mode)	Australia
Country of Birth of Parents (Mode)	Both parents born in Australia
Language Spoken at Home (Mode)	English
Ancestry 1st Response (Mode)	English
Social Marital Status (Mode)	Married in a registered marriage
Family Composition (Mode)	Couple family with children
Count of All Children in Family (Mode)	Two children in family
Highest Year of School Completed (Mode)	Year 12 or equivalent
Unpaid Domestic Work: Number of Hours (Mode)	5 to 14 hours
Number of Motor Vehicles (Mode)	Two vehicles
Number of Bedrooms in Private Dwelling (Mode)	Three bedrooms
Tenure Type (Dwelling Count) (Mode)	Owned with a mortgage

Q: How might you determine the distance between a given individual and Claire?

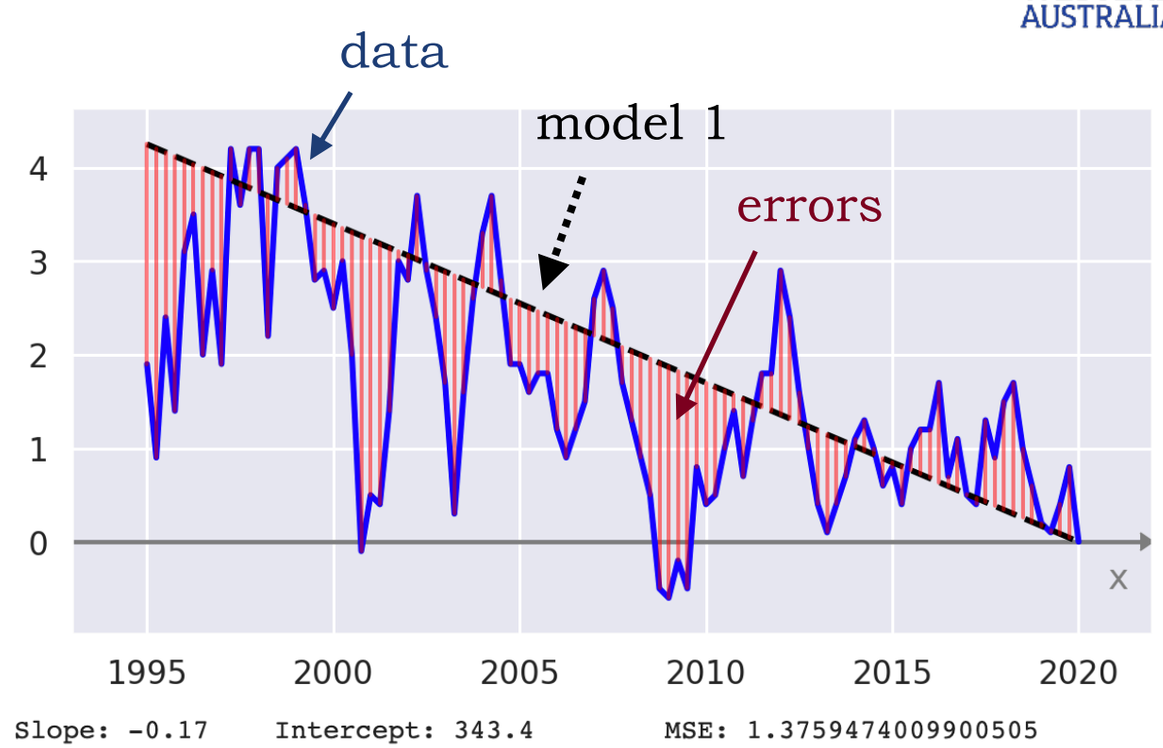
Is the Census a good model (abstraction) of a person?

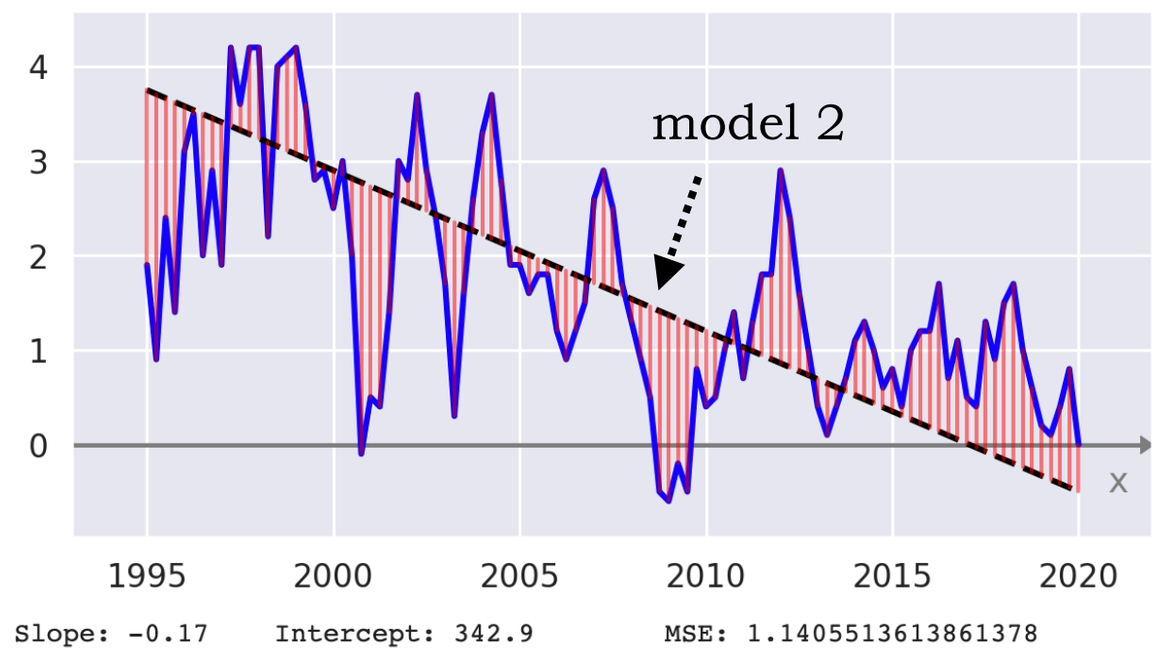
Numerical Features and Error

Tends to be easier in numerical domains…

- here features are numbers
- contribution to error is distance (difference) between model and sample at each x value
- eg. error is Euclidean distance
- expressed as a function…

$$\text{MSE} \equiv \frac{1}{N}\sum_{i=1}^{N}\left(f(x_i) - y_i \right)^2$$

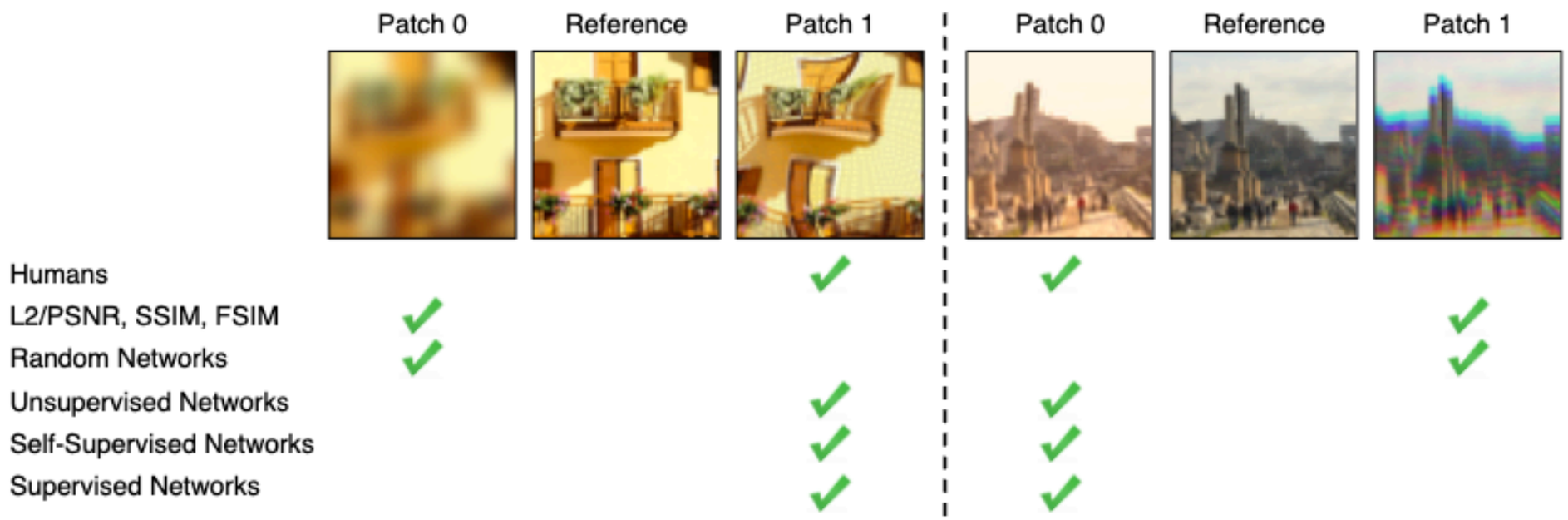




Numerical Features and Error

But beware!

Q: Which patch is "closer" to the middle patch?



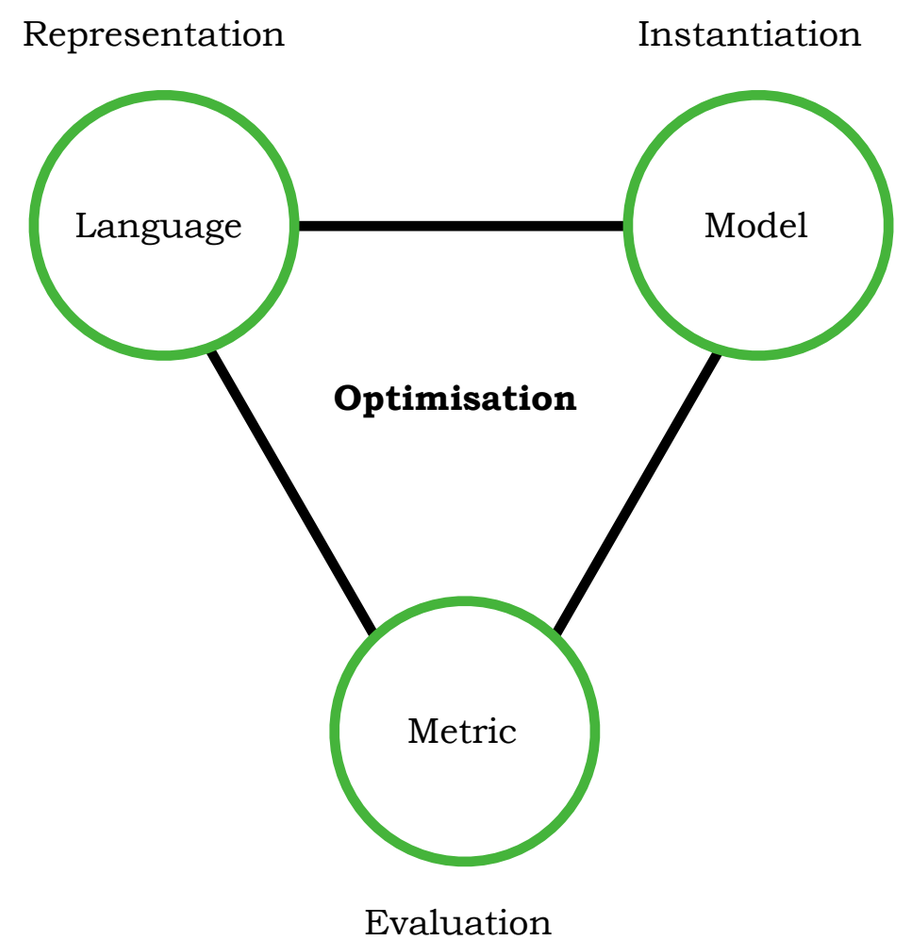
"The Unreasonable Effectiveness of Deep Features as a Perceptual Metric", Zhang et al, *CVPR 2018*.

Optimisation

We can now suggest a definition for optimisation

Ideal

- Find a model
- within the hypothesis space (determined by the language)
- that is *indistinguishable* from the target (zero error with respect to the selected features)

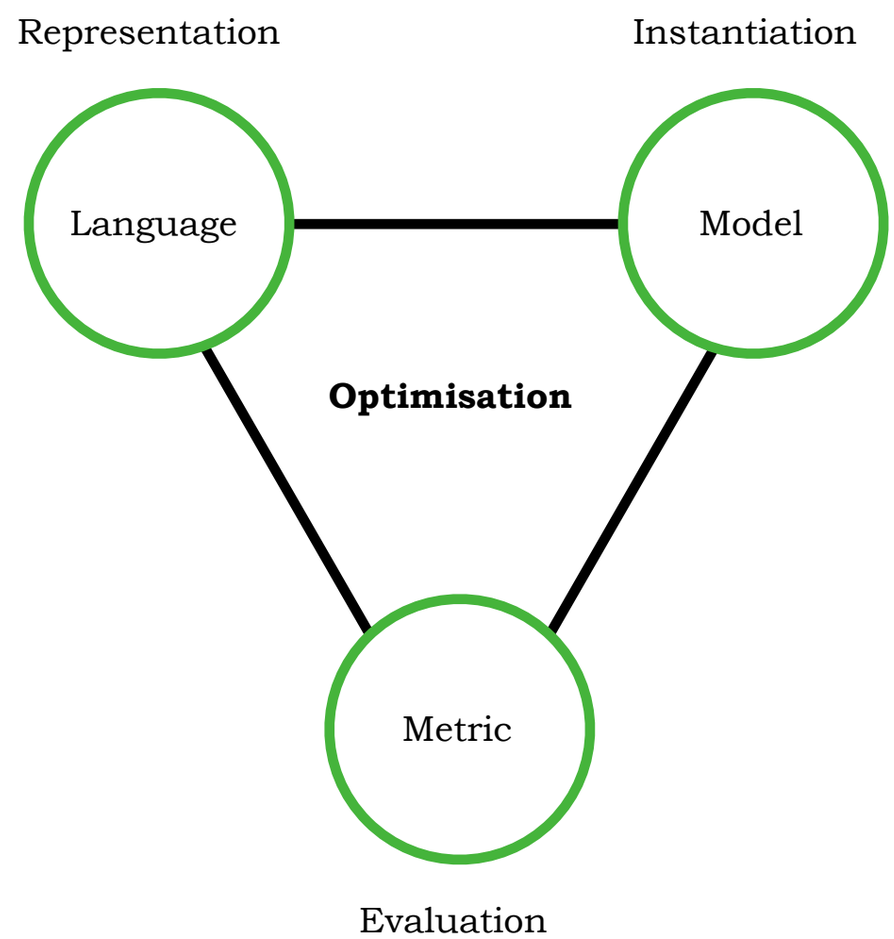


- Note 1: if the evaluated features are a “subset” of the description there could be many solutions - equivalent wrt the metric
- Note 2: in practice in complex problems its more likely there is no matching solution within the hypothesis space

Optimisation

Ideal 2

- Find a model
- within the hypothesis space (determined by the language)
- that is *closest* (minimal error) to the target

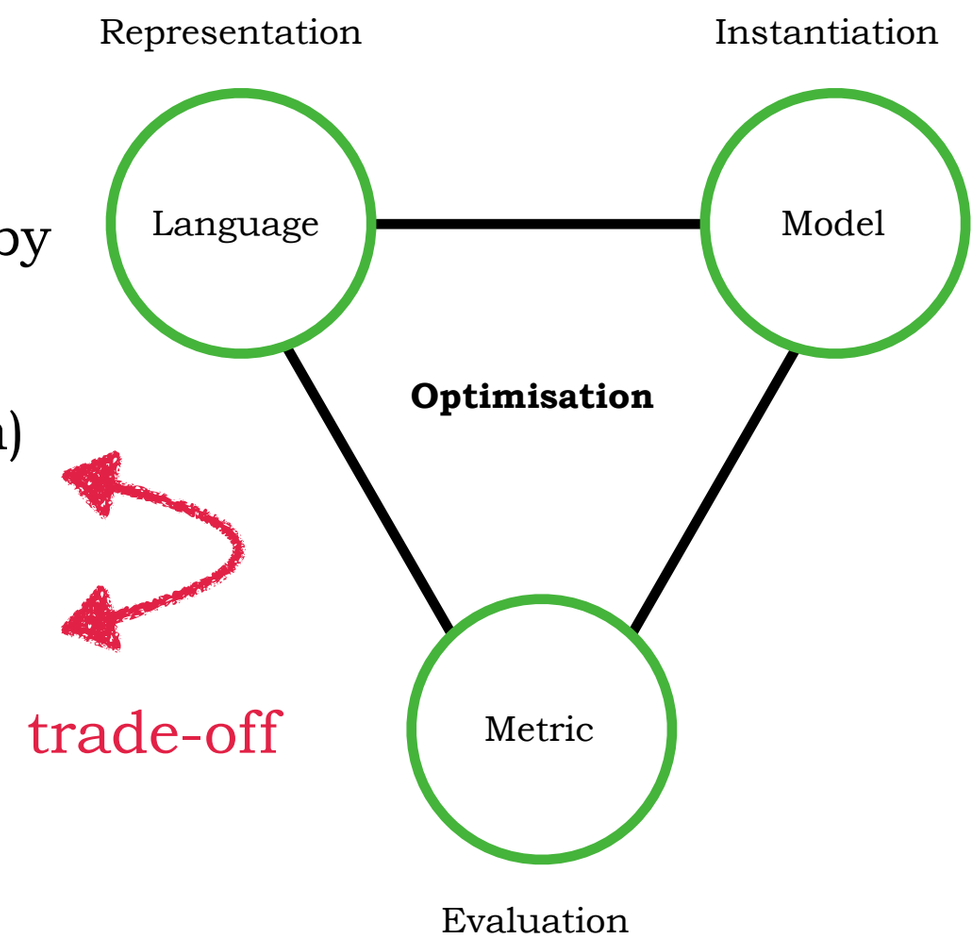


- Note 1: in complex problems it may be too expensive, or impossible (infinite domains) to find the best (closest) solution
- Note 2: sometimes the target may not be well defined (eg “infinite fitness”)

Optimisation

Practical definition

- Find a model
- within the hypothesis space (determined by the language)
- that is as ***close as possible*** (good enough) to the target
- within a ***specified amount of compute*** (time and space)



trade-off

- Distinguish between
  - ***online*** - compute in real time
  - ***offline*** - better result outweighs time cost

changes trade-off

- Next - lets see how these terms manifest in a concrete case study